Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/17/2025
PI Name:	Goree, John Ph.D.		
Project Title:	Nonlinear Wave Experiments in Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	06/14/2017	**End Date:**	12/14/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/14/2024 and new contract number is JPL 1689926 per J. Callas/JPL (Ed., 8/29/24) NOTE: End date changed to 12/31/2023 per J. Callas/JPL (Ed., 11/14/22) NOTE: End date changed to 12/13/2021 per U. Israelsson/JPL (Ed., 5/25/21) NOTE: Change in start and end dates to 6/14/2017-6/13/2021 (from 4/15/2017-4/15/2021) per I. Hahn/JPL and PI (Ed., 1/19/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1689926 ; JPL 1573629		
Performance Goal No.:			
Performance Goal Text:			

Description for Part 1 JPL 1573629 6/14/2017 to 12-14-2024
A flight experiment will be performed and analyzed in a study of nonlinear synchronization of waves. Synchronization is a cooperative phenomenon for oscillators or waves. A signature of synchronization is a frequency that shifts when an external modulation is applied. Compared to point-like oscillators, waves are more complicated due to their spatio-temporal behavior and dispersion properties. An understanding of synchronization is lacking for waves of all kinds, including those in dusty plasmas.

The experiment will use dusty plasmas, also called "complex plasmas." These are mixtures of small particles of solid matter, electrons, ions, and neutral gas. The small solid particles obtain large electric charges, so that they exhibit collective behavior. They are electrically confined, so that they do not contact any boundaries.

Microgravity enables this experiment by suppressing sedimentation, thereby allowing the waves to propagate distances of many wavelengths, with relatively uniform background conditions. This experiment is based on previous ground-based experiments, where waves propagated only a few wavelengths.

Our flight experiment will be performed using “Plasma Kristall 4” (PK-4), which is a facility in the Columbus Module of the International Space Station (ISS), built by the European Space Agency (ESA).

SCIENTIFIC/TECHNICAL MERITS

The main scientific topic is the synchronization of a dust acoustic wave (DAW). A DAW is a low-frequency compressional wave, which involves charge-separation from electrons and ions. The DAW is self-excited by an ion flow, and it grows naturally to a large amplitude so that nonlinear effects occur readily. Synchronization experiments are performed by applying electrical waveforms to an electrode to modulate the dusty plasma sinusoidally. The diagnostic for detecting the wave motion is video imaging using particle observation cameras. Image data will be analyzed to prepare space-time diagrams, which provide a spatial and temporal description of the wave's amplitude as it propagates. These space-time diagrams are the foundation for further analysis methods.

In support of our experiment, a theoretical effort will be undertaken at Iowa. The goal is to contribute to the development of a theory for synchronization of propagating waves, to compare to the experiment. The theoretical effort at Iowa will focus on numerical simulations using a hybrid model. Molecular dynamics simulation will also be performed. The main theoretical effort, which will center on analytic theory, will be done by theorist elsewhere, and we will collaborate with them.

The feasibility of our flight experiment is based on our previous ground-based experiments, which confirm that synchronization of the DAW occurs. These ground-based experiments also revealed unexplained phenomena, including a nonharmonic synchronization state, which we will investigate with our flight data.

In the unlikely event that our main experiment with waves does not yield useful results, we have a plan for a backup experiment to investigate mobility-limited motion of particles having a different size.

BROADER IMPACTS

This research has an overlap with nonlinear dynamics as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, a website, and Twitter. A K-12 outreach presentation will be made at an elementary school.

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Description for Part 2 JPL 1689926 12/15/2021-12/14/2024

Task Description:

Note: this extension was originally for two years, 12/15/21 - 12/14/23. It was followed by a one-year no-cost extension 12/15/23-12/14/24. The nature of the work changed, during this extension, as compared to Part 1, which was original proposal, as work on the PK-4 spaceflight mission was completed in 2021.

Activities during a two-year extension will focus on ground-based experiments and data analysis. These activities will be guided by an intention to enable future flight experiments. The scientific topics will be in the area of dusty plasmas, meaning small solid particles immersed in a partially ionized gas. There will be two research topics: static structure factor and afterglow.

The static structure factor S(k) is an inverse measure of particle positions, analogous to a diffraction pattern. The aim will be to determine an appropriate method of extrapolating S(k) measurements to small k values, because S(0) is a measure of the compressibility of the collection of particles. This will be done for particles in a disordered state with liquid-like characteristics, because such liquid states are commonly achieved in previous flight experiments with dusty plasmas.

The afterglow is made by switching off the power that sustains the plasma. During the afterglow, electrons and ions are lost to chamber walls within 2 milliseconds, but small solid particles remain in the volume afterwards. Under 1-g conditions these particles fall, but under microgravity conditions they would remain in the chamber for a much longer time. In our ground-based laboratory, we will measure the residual electric charge, after the surrounding plasma has vanished, and test theories for this charging. We expect that future microgravity flight experiments will allow afterglow experiments to study the same charging phenomenon, but over a much longer observation period.

Activities during the second year of a two-year extension will continue to be focused on ground-based experiments and data analysis. As before, these activities will be guided by an intention to enable future flight experiments. The scientific topics will be in the area of dusty plasmas, meaning small solid particles immersed in a partially ionized gas. In addition to the two research topics of the first-year extension, static structure factor and afterglow, there will now be the third topic of a shock wave.

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Shock experiment data will be analyzed using a new analysis method, for obtaining the nonequilibrium microscopic structure inside a shock layer. Shocks have been observed in previous microgravity dusty-plasma experiments, but their microscopic structure has not yet been studied.

Scientific reporting will be done in journals such as Physics of Plasmas, at conferences such as the Annual Meeting of the American Physical Society Division of Plasma Physics, and at a NASA workshop for Fundamental Physics.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

N/A

Task Progress:

The reporting period is 12/15/21 - 12/14/24.
This is Part 2 of the project, JPL/1689926. It includes two years of extension plus a final no-cost extension. During this period, all objectives were completed successfully.

The nature of the work done during this reporting period changed, as compared to Part 1, which focused on flight experiments on the PK-4 instrument on ISS. Work done during the present reporting period of 12/15/21 - 12/15/24 focused on ground-based experiments, data analysis, participation in the design of a future spaceflight mission (COMPACT), and reporting.

GROUND-BASED EXPERIMENTS

In anticipation of future microgravity experiments, we performed ground-based experiments and analyzed data from previous ground-based experiments. Flight experiments have a 3D suspension of particles that cannot be duplicated on the ground, where sedimentation causes particles to fall to a 2D layer. Ground-based experiments with the 2D layer establish the science and methods that will be used for 3D experiments under microgravity conditions.

Three kinds of experiments were performed and reported in publications. The first two kinds were proposed in the Statements of Work for award JPL/1689926, while the third kind is beyond what was proposed.

1. Ground-based experiments: viscosity of liquid-like dusty plasmas

One hallmark of dusty plasmas is that the micron-size particles can behave as macroscopic analogs of molecules in liquids or crystals. In these ground-based experiments, we established the measurement methods and parameter ranges that can be used in future spaceflight experiments, for example using the COMPACT instrument now being designed for the International Space Station. We focused our efforts on measurements of viscosity and measures of the hyperuniformity index.

This effort was completed successfully, resulting in three refereed papers in Physical Review E, numbered 1, 7, and 10 in the list below.

2. Ground-based afterglow experiments: charge measurements and charging processes Every experimental plasma must eventually be turned off, and when the power is switched off the result is what is called an afterglow, which lasts for at least a few milliseconds. We anticipate future microgravity experiments to follow up on the PI's 2003 Physical Review Letters paper that reported the first afterglow experiment under microgravity conditions, using the PKE instrument aboard the ISS. That first experiment led to many questions that could not be answered using the PKE instrument, so we performed ground-based experiments to establish the science and experimental hardware requirements for future microgravity experiments. This effort was completed successfully, resulting in six refereed papers, numbered 2,5, 6, 9, 11, and 12 in the list below.

3. Ground-based experiments for dust mitigation under Lunar conditions

Going beyond what we proposed, we also participated in a University of Colorado project to demonstrate a method of cleaning spacesuits, tools, and other surfaces on the Moon, in anticipation of future lunar habitation. This method uses an electron beam to charge surfaces under vacuum conditions, and make the dust jump off, without requiring any water or mechanical agitation.

This effort led to a refereed publication (number 4 in the list below) and a patent application.

PARTICIPATION IN THE DESIGN OF A FUTURE SPACEFLIGHT INSTRUMENT

The Principal Investigator (PI) was a member of the Science Definition Team (SDT) for the German/USA project called COMPACT, which is a future dusty plasma instrument for the ISS. The SDT had members from Germany, USA, and Canada, and it met by Zoom with representatives from the German Space Agency (DLR), its contractor (JPL), and the U.S. National Science Foundation.

This effort completed successfully in 2024, with the advancement of the project from one of establishing scientific requirements to an engineering phase. This effort also led to one refereed paper, number 4 in the list below.

The nature of the work done during this reporting period changed, as compared to experiments on the PK-4 spaceflight instrument during the periods ending 12/14/21. Work done during the present reporting period of 12/15/21 - 12/15/24 focused on ground-based experiments, data analysis, participation in the design of a future spaceflight mission (COMPACT), and reporting.

REPORTING

The PI had 12 peer-reviewed journal papers, 2 patent applications, 1 PhD dissertation, and 9 talks at various conferences and workshops. The latter included the 2023 and 2024 NASA Fundamental Physics Workshops.

[Ed. Note: Please see the Cumulative Bibliography for a comprehensive list of peer-reviewed publications with hyperlinks, invited talks/presentations, and patent information.]

Papers published in peer-reviewed journals, Dec 2021 - Dec 2024:

1. Jorge Berumen and J. Goree, Frequency-dependent complex viscosity obtained for a liquid two-dimensional dusty plasma experiment, Physical Review E, Vol. 105, art. no.015209 Jan 2022, DOI: 10.1103/PhysRevE.105.015209

2. Neeraj Chaubey and J. Goree, Preservation of a dust crystal as it falls in an afterglow plasma, Frontiers in Physics, Vol. 10, art. no. 879092 May 2022, DOI: https://

3. Christina A. Knapek, L. Couedel, A. Dove, J. Goree, Uwe Konopka, A. Melzer, S. Ratynskaia, Markus H. Thoma, Hubertus M. Thomas, COMPACT—a new complex plasma facility for the ISS, Plasma Physics and Controlled Fusion, Vol. 64, art. no. 124006, Nov 2022, DOI: 10.1088/1361-6587/ac9ff0

4. B. Farr, X. Wang, J. Goree, I. Hahn, U. Israelsson and M. Horányi, Dust removal from a variety of surface materials with multiple electron beams, Astro Astronautica, Vol. 200, pp. 42-47 Nov 2022, DOI: https://

5. Neeraj Chaubey and J. Goree, Coulomb expansion of a thin dust cloud observed experimentally under afterglow plasma conditions, Physics of Plasmas, Vol. 29, art. no. 113705 Nov 2022, https://

6. Job Beckers, Ranganathan Gopalakrishnan, John Goree, Particle interaction with afterglow plasma and non-quasi-neutral plasma, Frontiers in Physics, Vol. 10, p. 1167, May 2022, DOI: https://

7. Vitaliy Zhuravlyov, J. Goree, Jack F. Douglas, Paolo Elvati, and Angela Violi, Comparison of the static structure factor at long wavelengths for a dusty plasma liquid and other liquids, Physical Review E, Vol. 106, art. no. 055212, Nov 2022, DOI: 10.1103/PhysRevE.106.055212

8. Ajaz Mir, Sanat Tiwari, Abhijit Sen, Chris Crabtree, Gurudas Ganguli, and John Goree, Synchronization of dust acoustic waves in a forced Korteweg-de Vries-Burgers model, Physical Review E, Vol. 107, article no. 035202, Mar 2023, DOI: https://

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10. Vitaliy Zhuravlyov, John A Goree, Paolo Elvati and Angela Violi, Finite-size effects in the static structure factor S(k) and S(0) for a 2D Yukawa liquid, Physical Review E, Vol. 108, art. no. 035211, Sep 2023, DOI: https://

11. Neeraj Chaubey and J. Goree, Mitigating dust particle contamination in an afterglow plasma by controlled lifting with a DC electric field, Journal of Physics D, Vol. 57, art. no. 105201, Dec 2023, DOI: https://

12. Neeraj Chaubey and J. Goree, Controlling the charge of dust particles in an afterglow by modulating the plasma power, Journal of Physics D, Vol. No. 57, art. no. 205202, Feb 2024, DOI: https://

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